

# GeoDrillNet: Deep Learning for Continuous Wellbore Geology Prediction from Drilling Measurements

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## Abstract

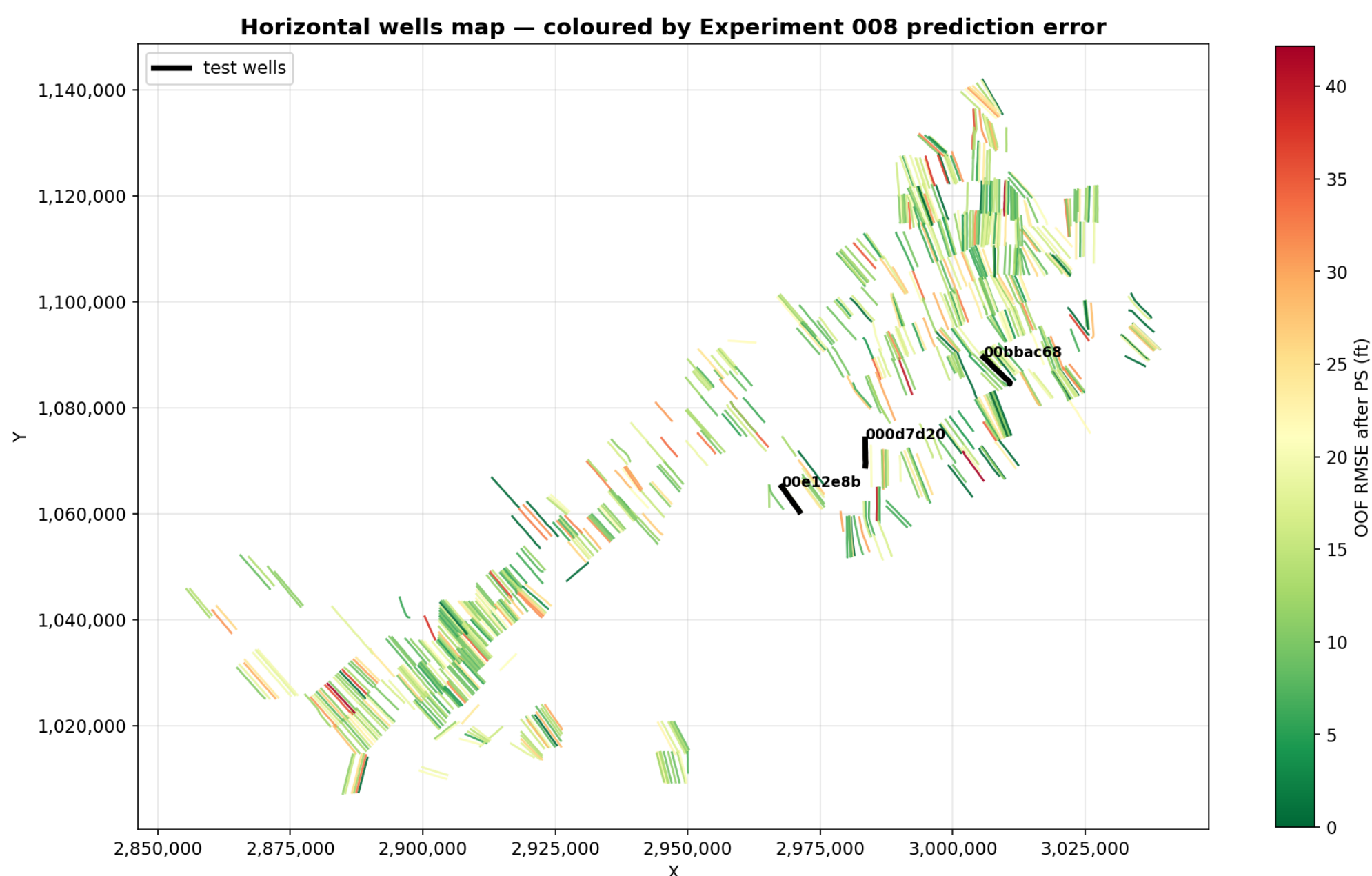
Continuous wellbore geology prediction is fundamental to automated geosteering during horizontal drilling, yet existing data-driven approaches often struggle to capture the complex spatial and temporal relationships present in drilling measurements. We propose **GeoDrillNet**, a deep learning framework that learns geological representations directly from sequential drilling data. GeoDrillNet combines hierarchical convolutional feature extraction with Transformer self-attention to characterize both local geological variations and long-range formation trends, predicting the geological position (TVT) along the well trajectory. Out-of-fold evaluation on 773 wells of the ROGII Wellbore Geology Prediction benchmark gives an RMSE of **21.33 ft** on the prediction interval, matching a larger TCN–cross-attention variant (21.40 ft) with **5× fewer parameters**.

## Motivation & Problem

- Geosteering keeps a horizontal well inside the target formation; the geological position (TVT) is picked manually by correlating gamma ray (GR) with a vertical typewell — slow, subjective, and hard to scale to real time.
- Target:** predict TVT beyond the **Prediction Start (PS)** point from the well path (X, Y, Z), GR, and the typewell log.
- Metric:**  $RMSE = \sqrt{\frac{1}{N} \sum_i (TVT_i^{\text{manual}} - TVT_i^{\text{pred}})^2}$  over predicted points.

## Dataset

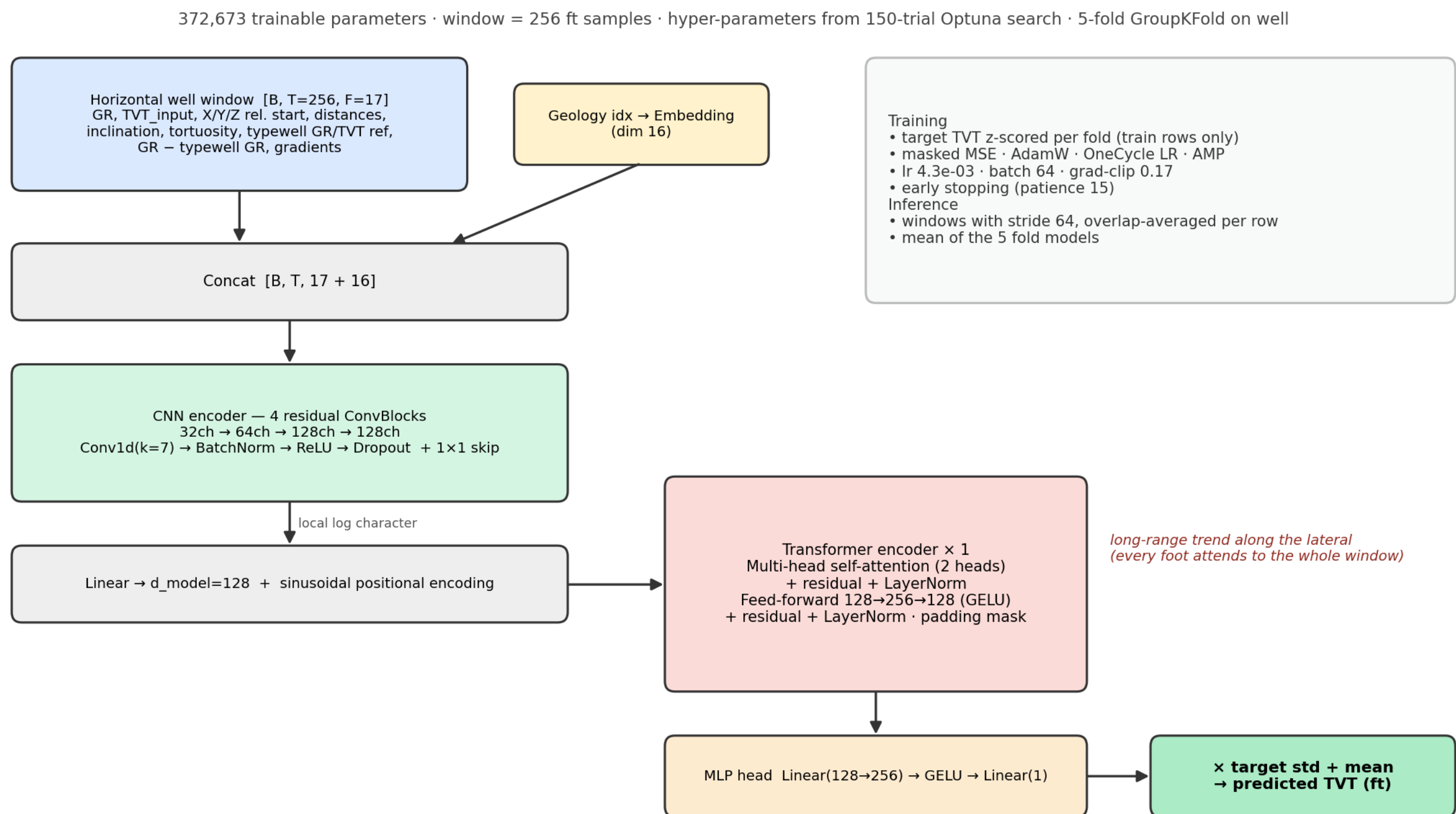
773 horizontal wells (one-foot sampling) with an assigned typewell each. Wells are split by 5-fold GroupKFold so every well is evaluated by a model that never saw it; colour shows each well's out-of-fold error.



## Proposed: GeoDrillNet (CNN + Transformer)

Residual 1D-CNN blocks capture local log character; a Transformer encoder relates every foot of the window to the whole lateral. Target standardized per fold; overlapping windows and 5 fold models are averaged.

Experiment 008 — CNN + Transformer (TVT regression, per-fold target standardization)

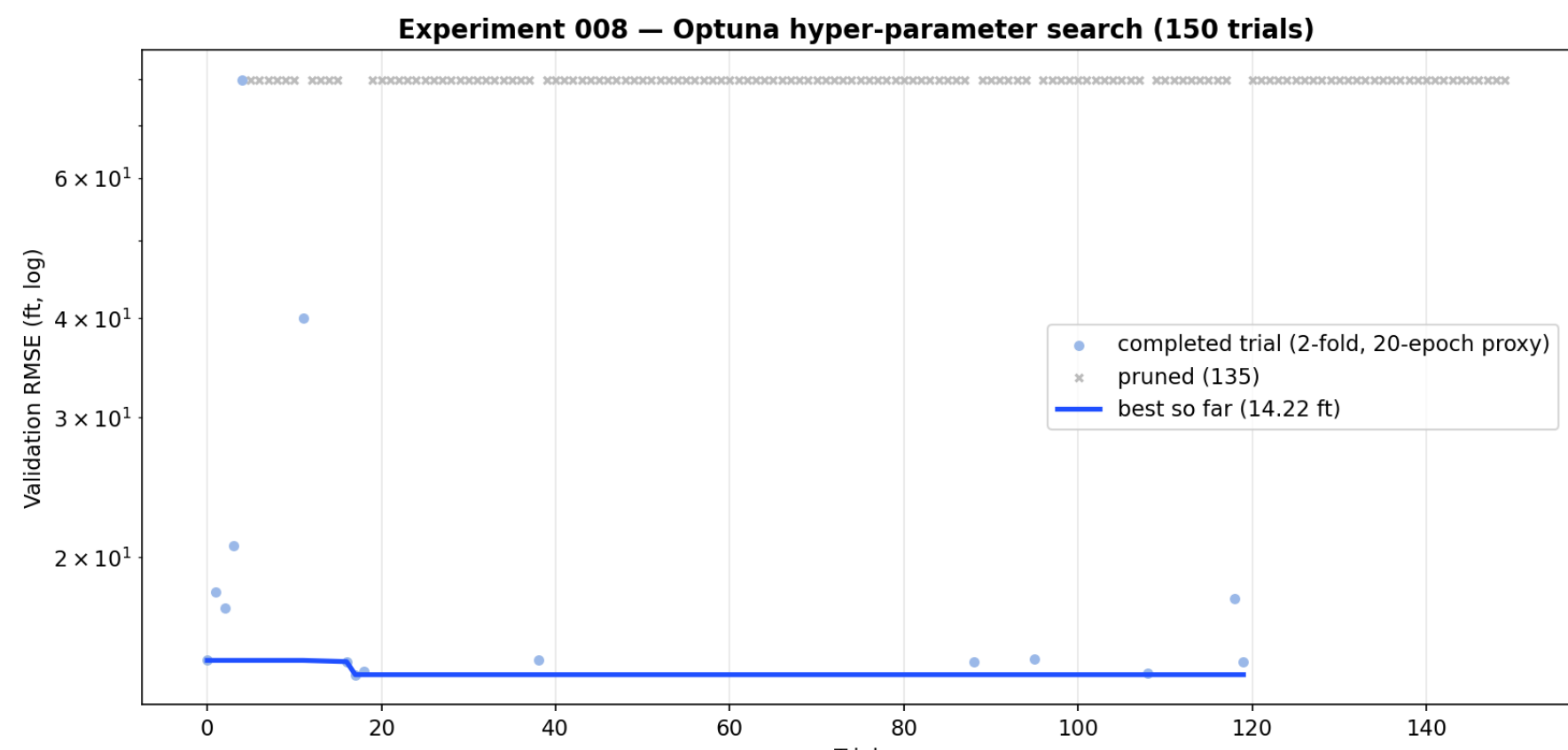


## Conclusions

- Efficiency:** 0.37M parameters, 5× fewer than the TCN–cross-attention model, at the same accuracy after PS (21.33 vs. 21.40 ft).
- Accuracy:** median per-well RMSE of 14.5 ft; the best held-out wells stay within 9–12 ft of the manual TVT over 2,500–5,500 ft of prediction.
- Limitation:** error grows with distance from PS; a constant-TVT baseline (15.91 ft) is still lower on the prediction interval.
- Next steps:** predict  $\Delta TVT$  relative to the last known value, explicit GR–typewell alignment, and stable mixed-precision training.

## Hyper-Parameter Search

150-trial Optuna search (TPE sampler, median pruning) over CNN/Transformer size, dropout, learning rate and window length.



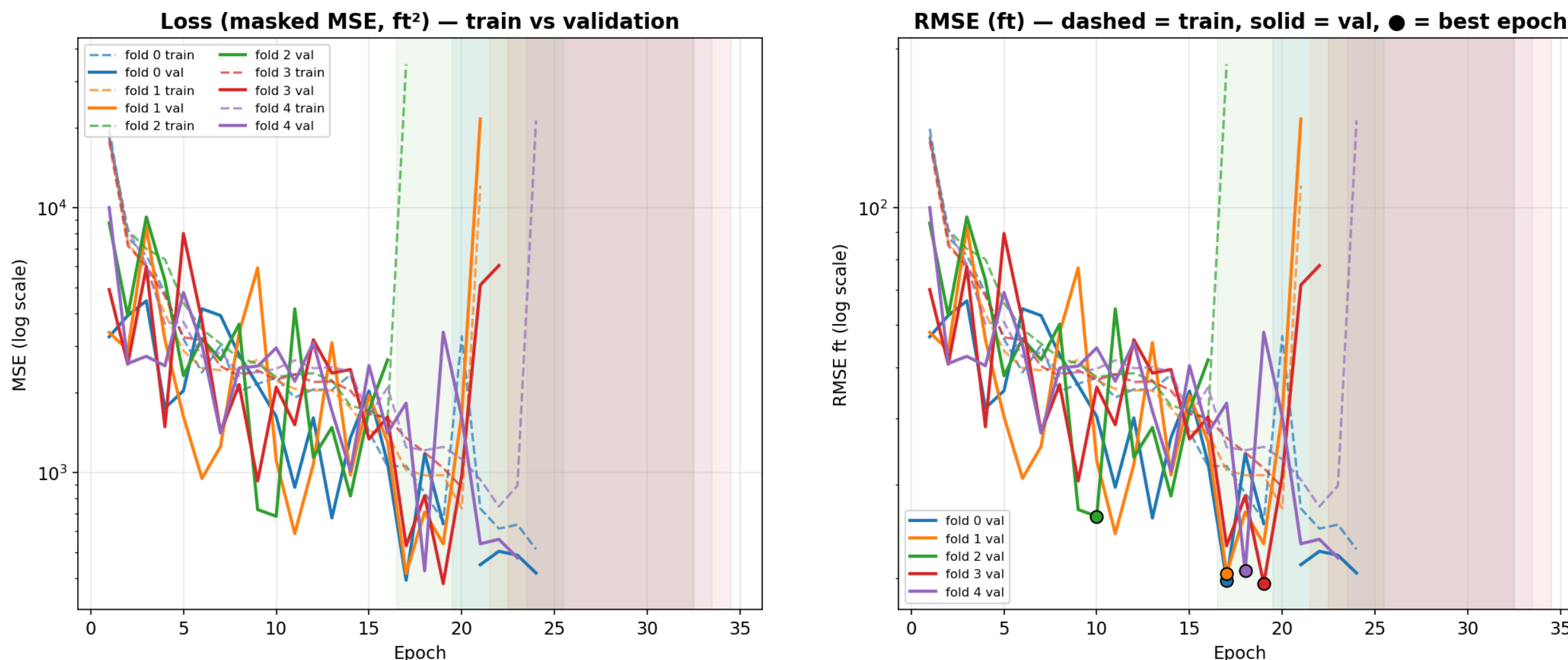
## Model Benchmarking

| Model                                  | Params       | Train Time   | RMSE after PS   |
|----------------------------------------|--------------|--------------|-----------------|
| <b>GeoDrillNet (CNN + Transformer)</b> | <b>0.37M</b> | <b>684 s</b> | <b>21.33 ft</b> |
| TCN + Cross-Attention + Transformer    | 1.91M        | 1238 s       | 21.40 ft        |
| Hold last known TVT constant           | —            | —            | 15.91 ft        |

Out-of-fold, 5-fold GroupKFold by well; train time = 5-fold CV on one GPU. A 51-ft moving average of GeoDrillNet's output gives 21.16 ft.

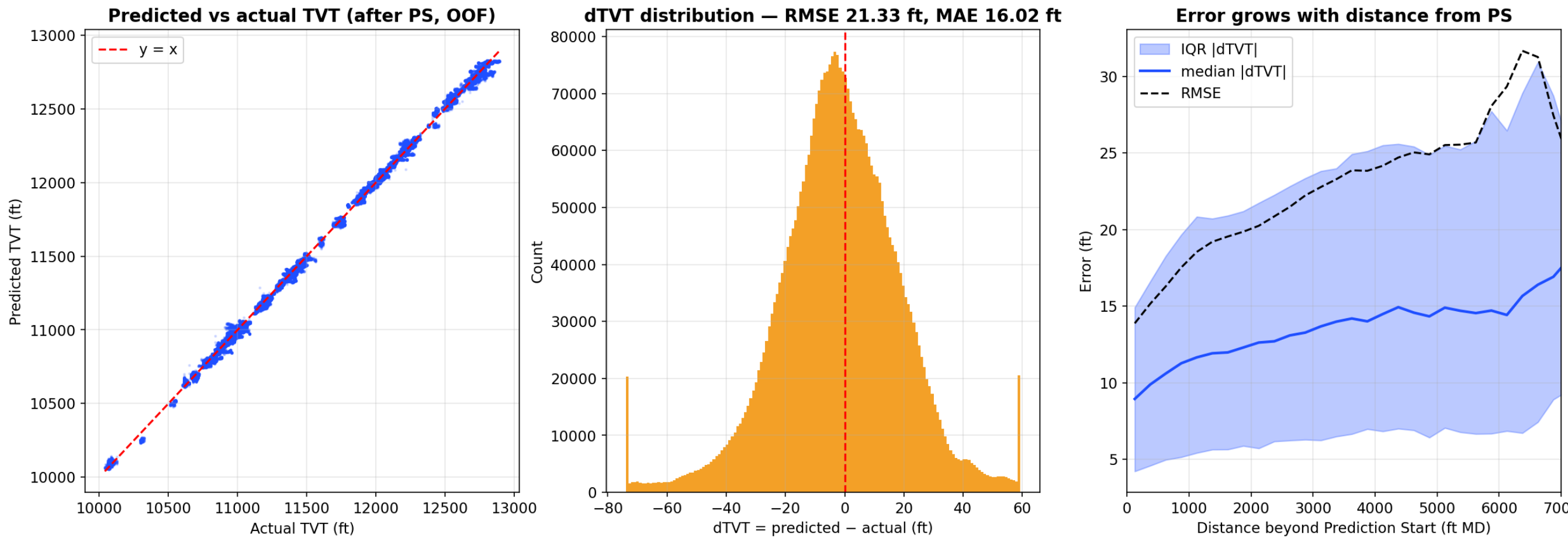
## Training Dynamics

Experiment 008 — Training Curves per Fold (shaded = epochs where loss became NaN; best weights from before divergence were kept)



## Prediction Quality (Out-of-Fold)

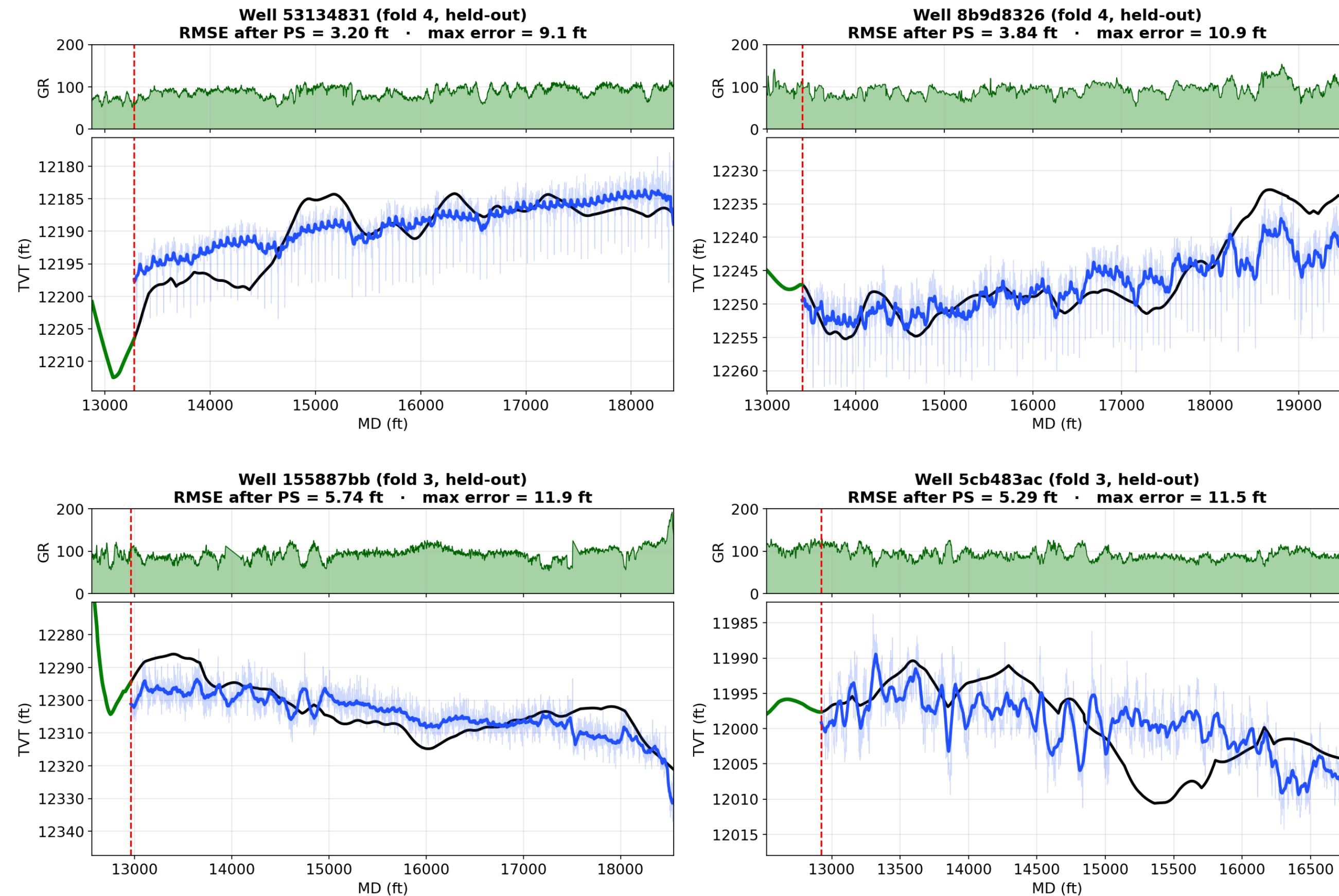
Experiment 008 — Prediction Quality (out-of-fold, all training wells)



## Held-Out Well Predictions

Wells never seen in training, predicted beyond PS over 2,500–5,500 ft: GeoDrillNet stays within 9–12 ft of the manual interpretation along the entire lateral (RMSE 3.2–5.7 ft).

GeoDrillNet — held-out wells with the smallest maximum error (>2,500 ft predicted, >20 ft TVT change)



— Actual TVT (manual interpretation) — Known TVT (before PS) — GeoDrillNet raw per-foot output — GeoDrillNet prediction (51-ft moving average) - - - Prediction Start

## Per-Well Error Distribution

